MILITARY SPECIFICATION

VISCOSIMETER, SAYBOLT; without Viscosity Tubes

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers a Saybolt Viscosimeter apparatus with four-tube capacity for determining the empirical measurement of Saybolt viscosity of petroleum products in accordance with procedures specified in ASTM D88 (see 2.2). The viscosity tubes are not included.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply to the extent specified herein. These documents may be obtained as directed by the contracting officer.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-M-7298	MANUAL, TECHNICAL: Commercial Equipment.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI	C73.11	Dimensions of Plugs and Receptacles-General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D88	Saybolt Viscosity, Test for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Description. The Saybolt Viscosimeter apparatus shall be a baked-on enameled steel or a type 18-8 corrosion-resisting steel housing a bath, flask shelf and electrical control compartments. The housing shall be designed to minimize loss of internal heat during operation. The bath housing shall serve as a container for the bath medium, and shall be supported with a chemical-resisting cover with four 2-1/4 inch Saybolt tube ports. The Saybolt tube ports shall be capable of vertically holding in position four Saybolt Viscosimeters that conform to requirements of ASTM D88. Means shall be provided for the lower ends of the inserted viscosimeters to be fastened to the bottom housing for the bath compartment. The bath compartment shall be supplied with a motor-driven stirring device, a cooling coil, heaters, and thermostatic sensing element. The heaters and coil shall be oriented with respect to the viscosimeters

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as indicated in ASTM D88. The shelf compartment shall be located beneath bath compartment to allow receiving flasks specified in ASTM D88 to collect effluent material during testing. The shelf shall have means for properly positioning the flasks beneath their viscosimeters. The base of the Saybolt Viscosimeter apparatus shall be stable and shall not rock when set on a flat surface. The overall dimensions of the apparatus, including projections, shall not exceed 26-1/2H x 14-3/4W x 17-7/8D inches; and the weight shall not exceed 50 pounds. The apparatus shall be supplied with a 3-conductor (with ground lead) line cord fitted with a 3-blade male plug that is suitable for use with receptacles specified in ANSI C73.11. The cord shall have a current carrying rating that is not less than the maximum current rating of the apparatus, and shall have an overall exposed length of at least 5 feet.

3.1.1 Accessories. The Saybolt Viscosimeter apparatus shall be furnished with the following accessories: Four chained corks, four thermometer supports, one pipet with rubber bulb, one oil strainer, one viscosity tube port closure, one viscosity tube cleaner, one tube nut wrench, and three gallons of white technical bath oil having the following characteristics: flash point 330° F., fire point 390° F., viscosity of 95 seconds S.U. at 100° F., color + 30 Saybolt, specific gravity 0.85 (60° F.).

3.2 Performance. The Saybolt Viscosimeter apparatus shall be designed to operate on 115-120 volts, 50-60 cycles AC with maximum power consumption of 1100 watts. The bath compartment shall be capable of being filled and drained with fluid medium, include a means of establishing a liquid level within the bath, and maintain the level of the bath medium to at least 1/4 inch above the level of the overflow rim of each inserted Saybolt Viscosimeter. The apparatus shall be capable of controlling and indicating the temperature of the bath medium at any temperature between 70° and 210 ° F. within limits specified in ASTM 88.

3.3 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298 MANUAL, TECHNICAL: Commercial Equipment. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

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3.4 Identification marking. The viscosimeter apparatus shall be legibly and permanently marked with the name or registered trademark of the manufacturer, Federal Stock Number, electrical ratings, contract or purchase order and "U.S.". Marking secured by adhesives shall not be permitted.

3.5 Workmanship. Workmanship shall be first class throughout. The viscosimeter apparatus shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the bath oil, may be furnished in lieu of actual testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

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Table I. For examination

	Inspection level	AQL (Percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Tests shall be conducted on all Saybolt Viscosimeter apparatuses.

4.3 Examination. The Saybolt Viscosimeter apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in table II.

Table II

Categories	Defects*
Major	
101	Apparatus not free of rocking when placed on level and smooth surfaces.
102	Apparatus not complete with all required components.
103	Line cord not firmly secured within housing of apparatus.
104	Line cord not free of cuts or tears.
105	Housing of apparatus not free of cracks.
106	Housing of apparatus not free of corrosion.
107	Plug on line cord not free of damage or not properly formed.
Minor	
201	Housing not free of sharp edges, pits or burrs.
202	Housing not free of dents.
203	Identification marking not present, complete, or legible.
204	Housing not free of dirt, grease or other contamination.

*Defects should not be limited to the above listed classifications.

4.3.2 Dimensional examination. The Saybolt Viscosimeter apparatus shall be examined for defects in dimensions. Any dimension not within limits specified shall be classified as a defect.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. An operational check shall be conducted to determine that the Saybolt Viscosimeter apparatus is capable of performing all significant functions.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One viscosimeter, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Each unit and its accessories shall be packaged in such a manner as to prevent marring, scratching and damage to the unit.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be overpacked in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Exterior container shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure and strapping shall be as specified in the appendix of the applicable box specification. Fiberboard exterior container shall conform to the special requirements as specified in PPP-B-636.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be overpacked as specified in 5.2.1.1, except domestic type containers shall be used. Case liner is not required.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Federal stock number (FSN).

6.2 This item is intended to cover the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6630-509-5254	VISCOSIMETER, SAYBOLT; without Viscosity Tubes

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6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6630-0186

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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Philadelphia, Pa. 19101

Defense Supply Agency

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

DOD-304

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-V-36845 (DSA-DM)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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